

Platform Layer & Workflow Engine - Estimation & Task Breakdown

Platform Layer, Workflow Engine & Tender Ingestion — Estimation & Task Breakdown

Companion to `plan-platform-layer-and-workflow-engine.md` — that document is the strategy/architecture; this one is the delivery estimate for the same phases.

Assumptions behind these numbers

These are **manual engineering estimates for a human team working at normal pace** — not AI-assisted generation time, and not “best case.” They assume:

- A team already familiar with this codebase’s conventions (the same team that’s been building Orelia CRM — no ramp-up time budgeted for learning `TenantOwnedEntity` /RBAC/audit patterns).
- **1 person-day = one developer, one working day**, inclusive of that task’s own dev-box testing, but **not** inclusive of code review, QA pass, staging verification, or bug-fix cycles — those are called out as separate line items per phase, not folded into each task silently.
- Backend and frontend tasks are listed separately; if backend and frontend developers work in parallel, **calendar time is roughly the longer of the two per phase**, not the sum — the “Person-days” column is total effort, not elapsed time. A rough calendar-time conversion assuming a 3-person team (2 backend, 1 frontend) working in parallel is given per phase.
- Phase 2 (Camunda), Phase 7 (tender scraping), and Phase 9 (RAG) carry the widest uncertainty bands in this whole document — new infrastructure/new dependency integration is inherently less predictable than extending existing patterns. Treat those

three phases' numbers as **±40%**, not the ±15% that applies to phases extending already-proven patterns (RBAC, audit log, dashboard widgets).

- Phase 7 numbers assume typical, moderately messy government/procurement tender-listing sites (some inconsistency in formatting, occasional PDF, no login wall). If the two actual Sri Lankan sites turn out to require login, heavy JS rendering, or scanned-image PDFs (needing OCR), add the contingency noted in that phase's row.
- Estimates exclude environment/DevOps overhead beyond what's explicitly listed (e.g. general CI pipeline maintenance, non-project infra work).

Phase 0 — Foundations

Task	Estimate	Notes
Add <code>FINANCE_*</code> / <code>LEGAL_*</code> / <code>HR_*</code> / <code>WORKFLOW_*</code> permission keys to <code>permissions.ts</code> + seed data	0.5 day	Mechanical, follows existing 4-permission pattern
Split HR permissions into tenant-scoped vs. <code>isPlatformOnly</code>	0.5 day	Small addition on top of the above
Design + migrate <code>tenant_module_config</code> table (<code>AuditedTenantEntity</code>)	1 day	New TypeORM entity + migration
Build <code>ModuleGuard</code> (mirrors <code>PermissionsGuard</code>)	1 day	New guard + unit tests
Code review + fixes	0.5 day	

Phase 0 subtotal: 3.5 person-days (~1 calendar week for 1 backend dev)

Phase 1 — On-demand module enabling

Task	Estimate	Notes
Backend: module-config CRUD service + controller	1.5 days	
Wire <code>AuditLogService.record()</code> into enable/disable	0.5 day	Reuses existing pattern

Task	Estimate	Notes
Apply ModuleGuard across module-scoped routes	1 day	Touches every route in scope, mechanical but broad
Frontend: System-tenant admin screen (toggle per tenant)	2 days	New page, table + toggle UI
Frontend: nav reads enabled-module list at layout load	1 day	
QA pass (enable/disable across 2–3 tenants, verify guard blocks disabled routes)	1 day	
Code review + fixes	0.5 day	

Phase 1 subtotal: 7.5 person-days (~1.5 calendar weeks, backend + frontend overlapping)

Phase 2 — Camunda infrastructure Δ wide band ($\pm 40\%$)

Task	Estimate	Notes
Add Zeebe + Operate to <code>docker-compose.yml</code> , verify local bring-up	1.5 days	New infra, first-time setup friction likely
Learning/spike: Zeebe client API, BPMN modeling basics	2 days	Real cost even for an experienced team if nobody has used Camunda 8 before — do not skip this line
Backend: Zeebe client wrapper module (start-instance, list-instances, complete-task)	3 days	
Backend: job-worker registration point for service tasks	2 days	
Author first BPMN process (Employee onboarding) in Camunda Modeler	2 days	Includes iterating the process model itself

Task	Estimate	Notes
Wire onboarding service tasks (provision accounts, assign certifications, manager checklist)	3 days	Touches employees / certifications / notifications
Integration testing (start → complete full onboarding instance end-to-end)	2 days	
Code review + fixes	1 day	

Phase 2 subtotal: 16.5 person-days (~2.5–3 calendar weeks) — treat as 10–23 days given the ±40% band

Phase 3 — Platform Finance/Legal/HR dashboards

Task	Estimate	Notes
Cross-tenant read endpoints — Finance (platform revenue/ARR rollups)	3 days	Scope depends on exact metrics wanted — see plan doc's open items
Cross-tenant read endpoints — Legal (shared templates/compliance)	2.5 days	
Super-tenant HR: reuse employees / certifications scoped to System tenant	1.5 days	Cheaper than Finance/Legal — no cross-tenant read, just new permission wiring
Dashboard widgets (3 modules) reusing existing permission-filtered widget framework	3 days	
Flag cross-tenant reads in api-endpoint-registry.md per the plan's data-residency note	0.5 day	Documentation task, not optional
QA pass (verify a non-System-tenant user cannot reach any of this)	1.5 days	Security-sensitive area — don't compress this

Task	Estimate	Notes
Code review + fixes	1 day	

Phase 3 subtotal: 13 person-days (~2 calendar weeks)

Phase 4 — Tenant-level Finance/Legal build-out

Scope here is genuinely open (the plan doc marks tenant-level Finance/Legal as “turn the Coming Soon placeholder into real features” without a finalized feature list yet) — the estimate below assumes a first, modest real feature per module, not a full accounting/legal suite. **Re-estimate this phase once the actual Finance/Legal feature list is confirmed — treat this row as a placeholder order-of-magnitude, not a committed number.**

Task	Estimate	Notes
Finance: first real feature (e.g. deal costing/invoicing view)	6 days	Backend + frontend combined, assumes 1 concrete feature
Legal: first real feature (e.g. tenant contract records)	5 days	Same caveat
RBAC + module-gating wiring for both	1.5 days	
QA pass	2 days	
Code review + fixes	1 day	

Phase 4 subtotal: 15.5 person-days (~2.5 calendar weeks) — treat as a placeholder pending real feature scoping

Phase 5 — Workflow UI + notification integration

Task	Estimate	Notes
Frontend: task inbox (“my tasks”) screen	2.5 days	
Frontend: process instance list + detail/history view	3 days	

Task	Estimate	Notes
Backend: endpoints backing both screens (calls into Phase 2's Zeebe wrapper)	2 days	
Wire task lifecycle events into modules/notifications	2 days	
QA pass	1.5 days	
Code review + fixes	1 day	

Phase 5 subtotal: 12 person-days (~2 calendar weeks)

Phase 6 — Additional workflow catalog rollout

Each row is one process: BPMN model + service-task wiring + minimal UI hook, at the same rigor as Phase 2's Employee onboarding reference build. **These can be reprioritized/reordered per the chief architect's call flagged in the main plan doc — the estimate per row doesn't change with order, only the total elapsed calendar time if done sequentially vs. in parallel.**

Process	Estimate	Notes
Recruitment (requisition → interview loop → offer)	6 days	Largest of this batch — most approval steps
Employee offboarding	3.5 days	Reuses onboarding's provisioning integrations in reverse
Deal approval (discount/terms threshold)	3.5 days	Touches existing Funnel/Deals stage-move logic
Tenant onboarding (internal provisioning checklist)	2.5 days	
Module-enable-request approval	2 days	Only needed if Phase 0's deferred plan-tier gating decision later requires an approval step
Contract/legal review	3 days	Depends on Phase 4's Legal module existing first

Process	Estimate	Notes
Expense/invoice approval	3 days	Depends on Phase 4's Finance module existing first
Integration testing across full catalog	2 days	
Code review + fixes (rolled up across catalog)	1.5 days	

Phase 6 subtotal: 27 person-days (~4–5 calendar weeks if largely sequential; less if multiple processes are built in parallel by different devs)

Phase 7 — Tender scraping + LLM extraction pipeline Δ wide band ($\pm 40\%$), site-details-pending

Task	Estimate	Notes
scraped_tender entity + migration	1 day	
Site A adapter (listing scrape + detail fetch)	3 days	Assumes server-rendered HTML, no login wall — add 2–3 days if it turns out to need a headless browser (JS-rendered pages), and add 3–4 more if tender detail is scanned-image PDF requiring OCR
Site B adapter	3 days	Same caveat as above — two sites are being estimated as roughly similar complexity, which may not hold
Daily @Cron job + dedup-by-externalRef logic	2 days	
LLM extraction integration (raw content → normalized fields)	3 days	Prompt design + structured-output parsing + retry/fallback handling
Debug logging per this project's standing logging rule	1 day	

Task	Estimate	Notes
QA pass against real site data (several real days of scrape runs, not just unit tests)	2.5 days	This genuinely needs to run against the live sites for multiple days to validate — budget calendar time, not just dev-days
Code review + fixes	1.5 days	

Phase 7 subtotal: 17 person-days (~3 calendar weeks) — treat as 12–24 days given the ±40% band and unresolved site-specifics

Phase 8 — Tender Suggestions UI + accept-to-Deal

Task	Estimate	Notes
Suggestions inbox page (list + filters)	2.5 days	
Accept flow: prefill <code>AddDealDialog</code> from <code>extracted</code> data	2.5 days	Reuses existing dialog, but wiring prefill + partial-field mapping takes real care
Reject/Ignore actions + status updates	1 day	
New <code>TENDER_SUGGESTIONS_VIEW / _UPDATE</code> permissions	0.5 day	
Sidebar badge/count	0.5 day	
QA pass (accept → verify Deal created correctly, audit log entry present)	1.5 days	
Code review + fixes	1 day	

Phase 8 subtotal: 9.5 person-days (~1.5–2 calendar weeks)

Phase 9 — RAG relevance ranking + smarter dedup (optional)

⚠ wide band ($\pm 40\%$)

Task	Estimate	Notes
Add pgvector extension + migration for embedding columns	1 day	
Embedding generation pipeline (new tenders + historical won deals)	3 days	Includes picking/wiring an embedding API
Relevance ranking in Suggestions UI (sort/score)	2 days	
Embedding-similarity dedup logic + “possible duplicate” flag in UI	3 days	
QA pass (verify ranking is sane against real historical deal data, not just synthetic)	2 days	
Code review + fixes	1 day	

Phase 9 subtotal: 12 person-days (~2 calendar weeks) — treat as 7–17 days given the $\pm 40\%$ band; lowest-priority phase in the whole plan, defer freely under time pressure

Phase 10 — Existing-system bug fixes & quality backlog

No dependency relationship with Phases 0–9 — this is already-shipped-system maintenance, not part of the platform/workflow initiative, and can be picked up independently of everything above, in any order.

Task	Estimate	Notes
Employee creation form: add missing required-field validations + fix employment-status dropdown options	1.5 days	Client + server-side validation, plus restricting the status enum options offered at creation
Prevent duplicate Contacts under the same Company	1.5 days	Pre-create uniqueness check (name/phone/email) + clear 4xx error; existing duplicate

Task	Estimate	Notes
		rows are a separate cleanup decision, not auto-merged
RBAC: auto-enable View when Create/Edit/Delete is granted (User Management, Teams, Roles, Employees)	2 days	<code>RolePermissionsDialog</code> UI logic + backend save-path enforcement (defense in depth, not UI-only)
Fix User Management status (Disable/Locked) not being enforced	1.5 days	Access-control gap — locate and fix the missing check in the auth/login path, then verify end-to-end
Fix Add Company → Contact tab country dropdown rendering	1 day	Scoped UI bug
Multiple sectors + multiple account tiers on Company/Customer creation	3.5 days	Data model change (single value → many-to-many with new join table), multi-select UI, migration for existing single values
Show phone (T.P.) number in Company → Contact view, not just name	1 day	Likely a response-shape/display gap rather than a missing column — confirm which before starting
QA pass across all seven fixes	1.5 days	
Code review + fixes	1 day	

Phase 10 subtotal: 14.5 person-days (~2.5 calendar weeks) — independent of the platform/workflow phases, can run in parallel with any of them

Grand total

Phase	Person-days	Rough calendar time (3-person team)
0 — Foundations	3.5	0.5–1 week
1 — Module enabling	7.5	1–1.5 weeks
2 — Camunda infra △	16.5 (band 10–23)	2.5–3.5 weeks

Phase	Person-days	Rough calendar time (3-person team)
3 — Platform dashboards	13	2 weeks
4 — Tenant Finance/Legal (placeholder)	15.5	2.5 weeks
5 — Workflow UI + notifications	12	2 weeks
6 — Workflow catalog rollout	27	4–5 weeks
7 — Tender scraping △	17 (band 12–24)	3 weeks
8 — Tender Suggestions UI	9.5	1.5–2 weeks
9 — RAG ranking/dedup △ (optional)	12 (band 7–17)	2 weeks
10 — Existing-system bug fixes	14.5	2.5 weeks (independent, can overlap any of the above)
Total (all phases)	~148.5 person-days	~23.5–26.5 calendar weeks if run mostly sequentially
Total excluding Phase 10 (existing-system bug fixes — separate workstream, not part of the platform/workflow initiative’s own scope)	~134 person-days	~21–24 calendar weeks

On sequencing for calendar time: the phases are not all dependencies of each other (see the mermaid dependency graph in the main plan doc) — Phase 4 can run parallel to Phase 3, Phase 7 can start as soon as Phase 1 lands without waiting on Camunda at all. A team large enough to run 2–3 phase-tracks in parallel could realistically compress the ~21–24 calendar-week sequential estimate down toward **14–16 calendar weeks**. The person-day totals above don’t change with parallelization — only calendar time does.